We want to solve the consumer's problem:

$$\max_{q(\omega)} Q = \left[\int_\Omega q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega\right]^{\frac{\sigma}{\sigma-1}} \text{ s.t.} \int_\Omega p(\omega)q(\omega)d\omega \leqslant E$$

This lets us set up the Lagrangian:

$$\mathcal{L} = \left[\int_\Omega q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega\right]^{\frac{\sigma}{\sigma-1}} + \lambda\left[E - \int_\Omega p(\omega)q(\omega)d\omega\right]$$

Now we take the first-order condition with respect to $q(\omega)$ and set it equal to zero. This gives

$$\cancel{\left(\frac{\sigma}{\sigma-1}\right)}\left[\int_\Omega q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega\right]^{\frac{1}{\sigma-1}} \cancel{\left(\frac{\sigma-1}{\sigma}\right)}\left(q(\omega)^{-\frac{1}{\sigma}}\right) - \lambda p(\omega) = 0$$

We can look at this first-order condition and recognize that the first surviving term is just $Q^{\frac{1}{\sigma}}$. Substituting this and rearranging to solve for λ, we get

$$\lambda = \frac{1}{p(\omega)} q(\omega)^{-\frac{1}{\sigma}} Q^{\frac{1}{\sigma}}$$

Now we consider a second good ω', which should be subject to the same logic mutatis mutandis. Both first-order conditions are rearranged to be set equal to λ, so we can write

$$\frac{1}{p(\omega)} q(\omega)^{-\frac{1}{\sigma}} \cancel{Q^{\frac{1}{\sigma}}} = \frac{1}{p(\omega')} q(\omega')^{-\frac{1}{\sigma}} \cancel{Q^{\frac{1}{\sigma}}}$$

From here it is simple to rearrange this equation to give

$$\frac{q(\omega)}{q(\omega')} = \left(\frac{p(\omega)}{p(\omega')}\right)^{-\sigma}$$

as required.

Now we want to derive the aggregate price index. We eventually do this by integrating the price function for ω over the goods index Ω, but we first need to do it in such a way that we get rid of $q(\omega)$. We do this by integrating out ω such that it becomes the aggregate

Q.

We start from our first-order condition rearranged for λ, and rearrange it to solve for $q(\omega)$:

$$\lambda = \frac{1}{p(\omega)} q(\omega)^{-\frac{1}{\sigma}} Q^{\frac{1}{\sigma}} \implies \lambda p(\omega) Q^{-\frac{1}{\sigma}} = q(\omega)^{-\frac{1}{\sigma}} \implies q(\omega) = \lambda^{-\sigma} p(\omega)^{-\sigma} Q$$

We now raise both sides to the power of $\frac{\sigma-1}{\sigma}$ to get

$$q(\omega)^{\frac{\sigma-1}{\sigma}} = \left[\lambda^{-\sigma} p(\omega)^{-\sigma} Q\right]^{\frac{\sigma-1}{\sigma}} = \lambda^{1-\sigma} p(\omega)^{1-\sigma} Q^{\frac{\sigma-1}{\sigma}}$$

Then we integrate:

$$Q^{\frac{\sigma-1}{\sigma}} = \int_\Omega q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega = \int_\Omega \lambda^{1-\sigma} p(\omega)^{1-\sigma} Q^{\frac{\sigma-1}{\sigma}} d\omega$$

We can pull the Q and λ terms out of the integral because they are independent of the quantity index ω, which gives us

$$\cancel{Q^{\frac{\sigma-1}{\sigma}}} = \cancel{Q^{\frac{\sigma-1}{\sigma}}} \lambda^{1-\sigma} \int_\Omega p(\omega)^{1-\sigma} d\omega \implies 1 = \lambda^{1-\sigma} \int_\Omega p(\omega)^{1-\sigma} d\omega$$

We now solve this for λ:

$$\lambda^{-(\sigma-1)} = \int_\Omega p(\omega)^{1-\sigma} d\omega \implies \lambda = \left[\int_\Omega p(\omega)^{1-\sigma} d\omega\right]^{-\frac{1}{1-\sigma}}$$

We can look at the resulting expression and recognize that it can be rewritten

$$\lambda = \left(\left[\int_\Omega p(\omega)^{1-\sigma} d\omega\right]^{\frac{1}{1-\sigma}}\right)^{-1} = \frac{1}{\left[\int_\Omega p(\omega)^{1-\sigma} d\omega\right]^{\frac{1}{1-\sigma}}}$$

The term in the denominator is recognizably a CES quantity, so let's call this the price index. Define it as

$$P \equiv \left(\int_{\Omega} p(\omega)^{1-\sigma} d\omega \right)^{\frac{1}{1-\sigma}}$$

so the equation becomes

$$\lambda = \frac{1}{P} = P^{-1}$$

Now, because we have integrated the original first-order condition, our solution for λ is now algebraically distinct from the original first-order condition. This allows us to substitute for λ in the (rearranged) first-order condition:

$$q(\omega) = \lambda^{-\sigma} p(\omega)^{-\sigma} Q = P^{\sigma} p(\omega)^{-\sigma} Q = \left(\frac{p(\omega)}{P} \right)^{-\sigma} Q$$

Now we derive a way to put expenditures E into this equation. Our equation for expenditures is the budget constraint $E = p(\omega)q(\omega)$, so we can get there by multiplying both sides by $p(\omega)$ and integrating:

$$E = \int_{\Omega} p(\omega) q(\omega) d\omega = \int_{\Omega} p(\omega) \left(\frac{p(\omega)}{P} \right)^{-\sigma} Q d\omega$$

Now we observe that the aggregate of prices, the bracketed expression P, will not depend on ω since we have already taken care of the ω by integrating over Ω! This lets us pull it (and Q) out of the integral:

$$E = \frac{Q}{P^{-\sigma}} \left(\int_{\Omega} p(\omega) p(\omega)^{-\sigma} d\omega \right) = \frac{Q}{P^{-\sigma}} \int_{\Omega} p(\omega)^{1-\sigma} d\omega$$

We recognize that the latter term is just $P^{1-\sigma}$, so we can finally write aggregate expenditures as

$$E = QP^{1-\sigma+\sigma} = PQ$$

which is what we want!

Now we observe that we obtained λ above as $\lambda = \frac{1}{P}$. This allows us to derive $q(\omega)$ as a

function of P from the first-order condition:

$$q(\omega) = \lambda^{-\sigma} p(\omega)^{-\sigma} Q = \left(\frac{p(\omega)}{P}\right)^{-\sigma} Q$$

Multiplying the right-hand side by $1 = \frac{P}{P}$, we derive an equation in terms of total expenditures E:

$$q(\omega) = \left(\frac{p(\omega)}{P}\right)^{-\sigma} \frac{PQ}{P} = \left(\frac{p(\omega)}{P}\right)^{-\sigma} \frac{E}{P}$$

as required.

Now we consider the problem of the monopolistically competitive plan. It chooses a price $p(\omega)$ to solve

$$\max_{p(\omega)} \pi = \left(p(\omega) - \frac{w}{\phi}\right) q(\omega)$$

where ϕ represents the idiosyncratic productivity of the firm. We now have an equation for $q(\omega)$, so we can substitute it to get

$$\max_{p(\omega)} \left(p(\omega) - \frac{w}{\phi}\right) \left(\frac{p(\omega)}{P}\right)^{-\sigma} \frac{E}{P} = \frac{E}{P}\left[\frac{p(\omega)^{1-\sigma}}{P} - \frac{w p(\omega)^{-\sigma}}{\phi P}\right]$$

We derive the first-order condition as

$$0 = \frac{E}{P}\left[(1-\sigma)\frac{p(\omega)^{-\sigma}}{P} + \frac{\sigma w p(\omega)^{-1-\sigma}}{\phi P}\right]$$

We can immediately move one term to the other side of the equation and get rid of $\frac{E}{P}$ and $\frac{1}{P}$, which appear in both terms. That gives

$$(\sigma - 1) p(\omega)^{-\sigma} = \sigma \frac{w}{\phi} p(\omega)^{-1-\sigma}$$

We can multiply both sides by $p(\omega)^{-1-\sigma}$ and get $p(\omega)$ on one side, so we do that to get the much better-looking

$$(\sigma - 1)p(\omega) = \frac{\sigma w}{\phi}$$

which becomes

$$p(\omega) = \frac{\sigma}{\sigma - 1}\frac{w}{\phi}$$

Now we have an equation for the price. Returning to the original firm problem, we maximized profits as

$$\pi = \left(p(\omega) - \frac{w}{\phi}\right) q(\omega)$$

We first consider just revenue, dropping the $-\frac{w}{\phi}q(\omega)$ term. This becomes

$$r(\phi) \equiv p(\omega)q(\omega) = \frac{\sigma}{\sigma - 1}\frac{w}{\phi}\left(\frac{p(\omega)}{P}\right)^{-\sigma}\frac{E}{P} = \left(\frac{\sigma}{\sigma - 1}\frac{w}{\phi}\right)p(\omega)^{-\sigma}EP^{\sigma-1}$$

We then substitute for $p(\omega)$ again:

$$r(\phi) = \left(\frac{\sigma}{\sigma - 1}\frac{w}{\phi}\right)\left(\frac{\sigma}{\sigma - 1}\frac{w}{\phi}\right)^{-\sigma}EP^{\sigma-1} = EP^{\sigma-1}\left(\frac{\sigma}{\sigma - 1}\frac{w}{\phi}\right)^{1-\sigma}$$

This is exactly the equation we want for revenue! Now we solve for profits:

$$\pi(\phi) = r(\phi) - \frac{w}{\phi}q(\omega) = EP^{\sigma-1}\left(\frac{\sigma}{\sigma - 1}\frac{w}{\phi}\right)^{1-\sigma} - \frac{w}{\phi}p(\omega)^{-\sigma}EP^{\sigma-1}$$

and again re-substitute for $p(\omega)$ to get

$$\pi(\phi) = EP^{\sigma-1}\left(\frac{\sigma}{\sigma - 1}\frac{w}{\phi}\right)^{1-\sigma} - \frac{w}{\phi}\left(\frac{\sigma}{\sigma - 1}\frac{w}{\phi}\right)^{-\sigma}EP^{\sigma-1}$$

We multiply the second term by $1 = \left(\frac{\sigma}{\sigma-1}\right)\left(\frac{\sigma-1}{\sigma}\right)$, which allows us to rewrite it as

$$\frac{\sigma-1}{\sigma}\left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)^{1-\sigma} E P^{\sigma-1}$$

Thus we can write profits as

$$\pi(\phi) = \left[E P^{\sigma-1}\left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)^{1-\sigma}\right]\left(\frac{\sigma}{\sigma}-\frac{\sigma-1}{\sigma}\right) = \frac{1}{\sigma}\left[E P^{\sigma-1}\left(\frac{\sigma}{\sigma-1}\frac{w}{\phi}\right)^{1-\sigma}\right] = \frac{r(\phi)}{\sigma}$$

as required.